Parte B - Data Quality Discussion

Source reconciliation and geographical coverage

Both sources use ISO3 codes, which served as the integration key; country names were explicitly checked for consistency. The supplied Maddison file covers 15 countries, whereas WDI and historical events also include the European Union (EUU). Consequently, EU warehouse records have NULL Maddison-sourced GDP, GDP per capita and population, while WDI measures are available within their respective coverage. The EU benchmark cannot support Maddison-based long-run comparisons, and must not be added to member-country totals.

Temporal coverage and missing observations

Maddison spans 1900–2022, while WDI generally begins around 1960, with indicator- and country-specific differences. GNI per capita starts globally in 1962, sector employment in 1991, and Portugal's sector value-added shares in 1995. Historical GDP-per-capita coverage also varies: Poland and Hungary have 34 and 23 internal missing years; the Baltic countries begin in 1973 with six internal gaps each; Czechia starts in 1970 without internal gaps; Ireland starts in 1913 with seven gaps. These counts concern gdppc, not necessarily every Maddison measure. No explicit NULL values were found in the profiled staging fields, but absent country–indicator–year combinations become missing measures after integration. NULLs were preserved rather than replaced with zero; COALESCE(..., 'NA') only made missingness visible in the report. Comparisons therefore require common observation windows and explicit treatment of incomplete decades.

Transformation and loading challenges

WDI's long format required mapping indicator codes and pivoting rows into country–year or country–year–sector records. Correct grouping and prior duplicate-key checks were essential to avoid merging unrelated observations. Sector and measure-type mappings aligned value-added and employment indicators; manufacturing employment remained NULL because no corresponding indicator was supplied. FULL OUTER joins preserved country–year records available in either or WDI. Insert / update enabled repeatable loads while retaining surrogate keys, but did not retain attribute history or automatically remove source deletions. Connectivity between containerized Saiku and host MySQL, database permissions, and conflicting Mondrian classes also required troubleshooting. Uneven series coverage complicated PRD charts; selecting aligned years improved readability but concealed annual variation. The report samples years at ten-year intervals, rather than calculating decade averages.

Main findings and interpretation limits

Portugal's relative position weakened against several Eastern European comparators: the decade-based results place it above all seven in the 1990s, but below Poland, Czechia, Hungary, Estonia and Lithuania in 2020–2022. These are comparisons of decade-average GDP per capita, not individual-year values. In Portugal, agriculture's GDP share fell from 4.8% to 2.1% between 1995 and 2020, while services rose from 58.5% to 65.6%. Industry and manufacturing shares also declined, but manufacturing is included within industry and cannot be added as an independent contribution. Falling shares do not necessarily imply falling absolute output. Group

averages are unweighted and reflect available observations; historical events provide context for turning points but do not establish their causes. These limitations qualify the findings without eliminating the evidence of long-run development and structural change.